Assignment 1 - Getting started

Sent 100%

28.1.2024

Forsøk 1

Se gjennom tilbakemelding
30.1.2024

Forsøk 1 Poengsum:
100%

Vis tilbakemelding

Anonym vurdering: **nei**

Ubegrenset antall forsøk tillatt
16.12.2023 til 29.4.2024

Detaljer

Introduction

This first assignment exists mostly to verify that the tools work as they should and for you to get familiar with the kit. You will also learn how to communicate between the microcontroller to the PC using serial communication, and to control an LED with a digital output.

NOTE: If Mbed Studio fails to upload your program to the board you might be using a charging cable, not a data cable. You can try a different cable before you ask for help.

Preparation

1. Complete **[Course preparations (https://uia.instructure.com/courses/14984/pages/course-preparations)](https://uia.instructure.com/courses/14984/pages/course-preparations)** if you haven't already. Every member of the group should do this.

2. Go to your *<course folder>* and create new folder *assignments*. You should avoid comitting and pushing large folders. Open Git Bash (Windows) or Terminal (OSX/Linux), go to *<course folder>/assignments* and enter the following command:

```
echo -e "BUILD\nmbed-os\n.mbed\n.github\n*.md" > .gitignore
```

3. Open Mbed Studio and select your *<course folder>/assignments* as workspace. Connect the development board to the PC or select target DISCO-L475VG-IOT01A (B-L475E-IOT01A). Create a new program based on example *mbed-os-example-blinky-baremetal* and name the program *assignment_1*. Now instead of storing the Mbed OS in the program folder, select link to the existing shared Mbed OS git repository you created during **[Course preparations (https://uia.instructure.com/courses/14984/pages/course-preparations)](https://uia.instructure.com/courses/14984/pages/course-preparations)** and add program. You should now have the first assignment located at *<course folder>/assignments/assignment_1*.

   If you rather select *mbed-os-example-blinky* in stead of the bare metal example, you will include lots of libraries for connectivity (BLE, Cellular, LoRaWAN, WiFi, NFC) and Internet protocols which we will not use at this stage, but the compile time will be huge (depending on of how powerfull your PC is). This can be avoided if you download file **mbedignore_no_network_connectivity.zip (https://uia.instructure.com/courses/14984/files/2350764?wrap=1)** **(https://uia.instructure.com/courses/14984/files/2350764/download?download_frd=1)** , unzip and add *.mbedignore* file next to your *main.cpp* file in the project.

4. Make sure *assignment_1* is selected as the active program. Build the program (press *the-blue-hammer-button*) and check the console output for errors. If everything is fine, upload the program (press *the-blue-play-button*) to the microcontroller and verify it is running OK (there should be one LED blinking on the development board).

Resources

- **LED resistor calculator (https://ledcalculator.net/)**
- **DigitalOut (https://os.mbed.com/docs/mbed-os/v6.16/apis/digitalout.html)**
- **BufferedSerial (https://os.mbed.com/docs/mbed-os/v6.16/apis/serial-uart-apis.html)**

Requirements

In this assignment you will create a program that lets you control an LED from the PC by pressing the 0 or 1 keys.

1. 25% Connect an LED (not blue) to an output with an appropriate resistor
2. 10% Configure the IO as output
3. 10% Configure the serial port with baud rate 115200
4. 10% Read 1 byte from the serial port:
   - 15% If the byte is '0' turn the led off
   - 15% If the byte is '1' turn the led on
   - 15% Do this forever

Testing

To test the program you must first upload it to the microcontroller. When the development board is connected to your PC, a *Serial Monitor* tab should be seen in the IDE. If not press *View* and select the *Serial Monitor*, then press a key to send that key to the microcontroller. If the printout is corrupt, make sure that the baud rate is set to 115200.

Delivery

Deliverer the following as text in Canvas:

- Link to Bitbucket where to code has been delivered.
  Directly to the file if there is only 1, or the directory if more than 1.
  Navigate to the delivery at **https://tools.uia.no/bitbucket (https://tools.uia.no/bitbucket)** and copy the URL from the browser.

- Link to YouTube (or other video service) with video showing each of the requirements. You don't need to show the actual code, just which of the requirements are met.